

RE-FLASHING SIGNA EXCITE SCP BOARD NVRAM IN THE FIELD

The SCP Board NVRAM can be re-flashed in the field. Before replacing the SCP try the steps below. If the battery measures below 2.5V DC it needs to be replaced. The battery a CR2477N that can be purchased at most Batteries Plus locations and Amazon.com.

***** Double-check your entries while going through step 4. Do not power cycle the MGD until you are sure that all the values are correct. If anything is mistyped the process will have to be restarted from the beginning. If you cycle the MGD power with a wrong NVRAM entry the SCP may hang. The only way to recover will be to clear the NVRAM by moving the jumper at J31 momentarily from the left position to the right position and then back again to the left position. The jumper must remain in the left, normal position during normal operation. *****

1. Open a C-shell window.
2. Login to the SCP Board by typing **bold** commands below.

```
-----  
t20 6# cu scp <Enter>  
Connected  
<Enter>  
EPPC-Bug>  
-----
```

3. If the **EPPC-Bug>** prompt is seen after logging in then the SCP Board NVRAM is corrupted. Proceed to step 4. Otherwise, the SCP Board NVRAM is **NOT** corrupted and the SCP Board is likely not the issue and the re-flash process below will not solve the problem.
4. Make the entries in **bold** print below.

```
-----  
EPPC-Bug>ENV;D <Enter>  
Update with Auto-Configuration Defaults  
  
Update Non-Volatile Memory (Y/N)? Y <Enter>  
  
Reset Local System (CPU) (Y/N)? N <Enter>  
WARNING: Updates will not be in effect until a RESET is performed.  
EPPC-Bug>ENV <Enter>  
Probe System for Supported I/O Controllers [Y/N] = Y? <Enter>  
Local SCSI Bus Reset on Debugger Startup [Y/N] = N? <Enter>  
PCI Interrupts Route Control Registers (PIRQ0/1/2/3) = 0A0B0E0F? <Enter>  
Firmware Command Buffer Offset = 000002C8? <Enter>  
Firmware Command Buffer Size = 00000200? <Enter>  
Firmware Command Buffer Delay = 5000? <Enter>  
Program Intermediate Load Address = 00200000? <Enter>  
Binary Program Load Address = 00080000? <Enter>  
Binary Program Execution Offset = 00000100? <Enter>  
Primary Network Controller LUN = 20? <Enter>  
Primary Network Device LUN = 00? <Enter>
```

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```
Firmware Command Buffer :  
['NULL' terminates entry]? GO FC000100 <Enter>  
SD <Enter>  
SE;AEM;QST;PCIB;FECETH <Enter>  
GO FC00017C <Enter>  
NULL <Enter>
```

```
Update Non-Volatile Memory (Y/N)? Y <Enter>
```

```
Reset Local System (CPU) (Y/N)? N <Enter>  
WARNING: Updates will not be in effect until a RESET is performed.  
EPPC-Bug>  
-----
```

5. The safest way to get the SCP Board out of debug mode is to cycle power at the back of the MGD Chassis.
6. Leave the MGD Chassis off for about 5 seconds before turning it back on.
7. Perform a **TPS Reset**.

*** If the C-shell window is open, and there is still a connection to the SCP Board, you should see information scrolling down the screen during the TPS Reset.

8. If the TPS Reset was successful then close the C-shell.

*** Once the TPS Reset is successful, perform a scan to confirm that the system is operating correctly. If it is not successful and the SCP Board is back in debug mode, repeat the process outlined in step 4 again. If the SCP Board still fails then power the MGD Chassis off and pull the SCP Board partially out of the slot as shown in Illustration 1. Locate the jumper J31 and the button battery on the bottom, front of the board.

- Confirm that jumper J31 is across the 2 pins closest to the front faceplate of the board. This is the normal position for J31.
- Confirm that the button battery is firmly seated in its holder. Use a digital multimeter to measure the battery voltage (should measure about 3.1 VDC and not less than 2.5 VDC) and confirm that it is not too low. The exposed, flat side of the battery is the (+) terminal while the side of the battery is the (–) terminal.
- Make certain that there are not any bent pins on the MGD Chassis mid-plane in the slot that contains the SCP Board or, for that matter, in any other slot.

When finished inspecting the SCP Board reseat it back into the MGD Chassis slot. If the NVRAM was cleared then repeat the process in step 4 again to re-flash the NVRAM.

If the jumper is in the correct position, the battery is seated properly, and the battery voltage is OK but the SCP Board still won't download, then replace it.